

Assignment 2

- 1. Explain the working of the Data Encryption Standard (DES). What are the key components of DES, and how does it ensure data security? Discuss the limitations of DES and why it was eventually replaced.**
(10 marks)
- 2. Describe the structure and working of the Advanced Encryption Standard (AES). How does AES improve upon DES in terms of security and performance? Explain the key steps involved in the AES encryption process.**
(10 marks)
- 3. Compare and contrast the Blowfish and IDEA algorithms. What are the key features of each algorithm, and how do they contribute to secure symmetric encryption? Provide examples of real-world applications for these algorithms.**
(10 marks)
- 4. What is the importance of confidentiality in cryptography? Explain the concepts of link encryption and end-to-end encryption. How does symmetric encryption ensure confidentiality, and what are the challenges in key distribution?**
(10 marks)
- 5. Describe the RSA algorithm in detail, including key generation, encryption, and decryption. Use a simple example to illustrate the process.**
(10 marks)
- 6. Explain the Diffie-Hellman key exchange protocol. How does it enable secure key exchange over an insecure channel? Provide a simple example to illustrate the process.**
(10 marks)